

MAT 535 HW04

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Problem 1 (§13.1#2). Show that $x^3 - 2x - 2$ is irreducible over $\mathbb{Q}$ and let θ be a root. Compute $(1 + \theta)(1 + \theta + \theta^2)$ and $\frac{1 + \theta}{1 + \theta + \theta^2}$ in $\mathbb{Q}(\theta)$.

Solution. If $x^3 - 2x - 2$ factors over $\mathbb{Q}$, it must have a linear factor, implying that it has a rational root. The rational root theorem says that a rational root of the given polynomial must be one of ± 1 or ± 2 , but we can check manually that none of these are roots (± 1 fail because the value of the polynomial would be odd, while ± 2 fail because the value of the polynomial would be $2 \pmod{4}$).

We can compute that

$$(1 + \theta)(1 + \theta + \theta^2) = 1 + 2\theta + 2\theta^2 + \theta^3 = \boxed{3 + 4\theta + 2\theta^2}$$

and

$$\frac{1 + \theta}{1 + \theta + \theta^2} = (1 + \theta) \cdot \frac{1}{3}(5 + \theta - 2\theta^2) = \frac{1}{3}(5 + 6\theta - \theta^2 - 2\theta^3) = \boxed{\frac{1}{3} + \frac{2}{3}\theta - \frac{1}{3}\theta^2}$$

□

Problem 2 (§13.2#4). Determine the degree over $\mathbb{Q}$ of $2 + \sqrt{3}$ and of $1 + \sqrt[3]{2} + \sqrt[3]{4}$.

Solution. The degree of $2 + \sqrt{3}$ is $\boxed{2}$; clearly the degree is greater than 1 because $\sqrt{3}$, hence $2 + \sqrt{3}$, is not rational, and the degree is at most 2 because $2 + \sqrt{3}$ is a root of $x^2 - 4x + 1$.

The degree of $\alpha = 1 + \sqrt[3]{2} + \sqrt[3]{4}$ is $\boxed{3}$. Obviously $\alpha \in \mathbb{Q}(\sqrt[3]{2})$, hence $\mathbb{Q}(\alpha) \subseteq \mathbb{Q}(\sqrt[3]{2})$, and we can compute

$$\alpha(\sqrt[3]{2} - 1) = (\sqrt[3]{2})^3 - 1 = 1 \iff \frac{1}{\alpha} + 1 = \sqrt[3]{2},$$

so the reverse inclusion $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{Q}(\alpha)$ also holds. Thus $\sqrt[3]{2}$ has the same degree as α , namely 3. □

Problem 3 (§13.2#8). Let F be a field with characteristic not equal to 2. Let D_1 and D_2 be elements of F , neither of which is a square in F . Prove that $F(\sqrt{D_1}, \sqrt{D_2})$ is of degree 4 over F if $D_1 D_2$ is not a square in F and is of degree 2 over F otherwise.

Solution. We show that $F(\sqrt{D_1}) = F(\sqrt{D_2})$ if and only if $D_1 D_2$ is a square in F .

First, if $D_1 D_2$ is a square in F , then there exists $a \in F$ such that $\sqrt{D_1} \sqrt{D_2} = a$, so $\sqrt{D_1} = a/\sqrt{D_2} \in F(\sqrt{D_2})$ and $\sqrt{D_2} = a/\sqrt{D_1} \in F(\sqrt{D_1})$ implies that $F(\sqrt{D_1})$ and $F(\sqrt{D_2})$ contain each other.

Conversely, if $F(\sqrt{D_1}) = F(\sqrt{D_2})$, then $\sqrt{D_1} = a_1 + a_2 \sqrt{D_2}$ for $a_1, a_2 \in F$. Squaring,

$$D_1 = (a_1^2 + D_2 a_2^2) + 2a_1 a_2 \sqrt{D_2}$$

implying that $2a_1 a_2 \sqrt{D_2} \in F$, hence $2a_1 a_2 = 0$. Since F does not have characteristic 2, either $a_1 = 0$, in which case $\sqrt{D_1} = a_2 \sqrt{D_2}$ implies that $D_1 D_2 = (a_2 D_2)^2$ is a square, or $a_2 = 0$, in which case $\sqrt{D_1} = a_1 \in F$, which contradicts the fact that D_1 is not a square; thus it must be the case that $D_1 D_2$ is a square in F .

Thus if $D_1 D_2$ is a square then $F(\sqrt{D_1}, \sqrt{D_2}) = F(\sqrt{D_1})$ has degree 2. If $D_1 D_2$ is not a square then

- (1) $F(\sqrt{D_1}, \sqrt{D_2})$ has degree divisible by the least common multiple of $[F(\sqrt{D_1}) : F]$ and $[F(\sqrt{D_2}) : F]$, namely 2.
- (2) $F(\sqrt{D_1}, \sqrt{D_2})$ has degree greater than 2, since $F(\sqrt{D_1})$ is a proper subfield of degree 2.

(3) $F(\sqrt{D_1}, \sqrt{D_2})$ has degree at most $[F(\sqrt{D_1}) : F][F(\sqrt{D_2}) : F] = 4$.

These three pieces of numerical information imply that $[F(\sqrt{D_1}, \sqrt{D_2}) : F] = 4$. $\square$

Problem 4 (§13.2#16). Let K/F be an algebraic extension and let R be a ring contained in K containing F . Show that R is a subfield of K containing F .

Solution. For any $\alpha \in R$, if $p(x) = c_0 + c_1x + \cdots + c_nx^n \in F[x]$ is the minimal polynomial of α over F (which exists because α is algebraic over F), then we can write

$$\alpha^{-1} = -c_0^{-1}(c_1 + c_2\alpha + \cdots + c_n\alpha^{n-1}).$$

Since R contains F and α it must contain the coefficients $-c_0^{-1}c_k \in F$ and the powers of α , so R contains α^{-1} for every $\alpha \in R$, hence R is a field contained in K and containing F . $\square$

Problem 5 (§13.2#20). Show that if the matrix of the linear transformation “multiplication by α ” considered in the previous exercise is A then α is a root of the characteristic polynomial of A . This gives an effective procedure for determining an equation of degree n satisfied by an element α in an extension of F of degree n . Use this procedure to obtain the monic polynomial of degree 3 satisfied by $\sqrt[3]{2}$ and by $1 + \sqrt[3]{2} + \sqrt[3]{4}$.

Solution. Suppose that $[F(\alpha) : F] = d$ and $[K : F(\alpha)] = n/d$. The extension $F(\alpha)/F$ has F -basis $1, \alpha, \dots, \alpha^{d-1}$, and $K/F(\alpha)$ has some $F(\alpha)$ -basis $\beta_1, \dots, \beta_{n/d}$, so that the elements $\alpha^j\beta_k \in K$ for $0 \leq j < d$ and $1 \leq k \leq n/d$ form an F -basis of K . Multiplication by α sends $\alpha^j\beta_k$ to $\alpha^{j+1}\beta_k$ for $0 \leq j < d-1$ and sends $\alpha^d\beta_k$ to $(-c_0 - c_1\alpha - \cdots - c_{d-1}\alpha^{d-1})\beta_k$ where $p(x) = c_0 + c_1x + \cdots + c_dx^d \in F[x]$ is the minimal polynomial of α over x , so the matrix of α on the invariant subspace generated by $\beta_k, \alpha\beta_k, \dots, \alpha^{d-1}\beta_k$ is

$$\begin{pmatrix} 0 & 0 & 0 & \cdots & -c_0 \\ 1 & 0 & 0 & \cdots & -c_1 \\ 0 & 1 & 0 & \cdots & -c_1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -c_{d-1} \end{pmatrix},$$

the companion matrix $M_{p(x)}$ of $p(x)$. Then the matrix of A with respect to our basis is

$$\begin{pmatrix} M_{p(x)} & 0 & \cdots & 0 \\ 0 & M_{p(x)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & M_{p(x)} \end{pmatrix}$$

where the 0s are $d \times d$ blocks of zeroes and $M_{p(x)}$ appears n/d times. The characteristic polynomial of this matrix is $p(x)^{n/d}$, which of course has α as a root.

Observe that $\alpha = \sqrt[3]{2}$ and $\beta = 1 + \sqrt[3]{2} + \sqrt[3]{4}$ are both in $F(\sqrt[3]{2})$, which has basis $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$. With respect to this basis, multiplication by α has matrix

$$\begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

which has characteristic polynomial $\boxed{x^3 - 2}$; multiplication by β has matrix

$$\begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

which has characteristic polynomial $\boxed{x^3 - 3x^2 - 3x - 1}$. $\square$